

What is Software-Defined Networking (SDN) Application?

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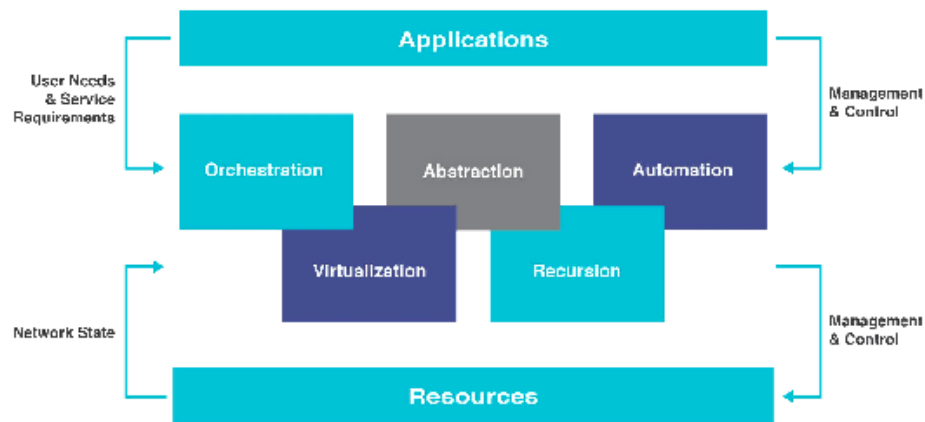
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All You Need To Know About SDN Application

What is Software-Defined Networking (SDN) Application?

The Architecture of Software-Defined Networks



Software-defined networking (SDN) application is a software program which is designed to perform a task in a software-defined networking environment. It is that approach to computer networking that not only allows network administrators to change programmatically, control, initialize, and manage network behaviour dynamically through open interfaces but also provides the concept of lower-level functionality. SDN applications also help in enlarging and substituting upon functions that are accomplished in the hardware devices of a regular network through firmware.

SDN could influence as much as [40% of all networking spending](#) within the next few years – SDN Market Size Expected to Reach \$35B by 2018,” SDN Central, April 24, 2013

Top networking initiatives in 2015



SOURCE: TECHTARGET 2015 PURCHASING INTENTIONS SURVEY

With the help of SDN, the administrators are able to manage and control the entire network through the abstraction of higher-level functionality. Even though management is abstracted, there has to be some kind of communication between the two planes, i.e. the control plane and the data plane.

Control plane – This system manages how traffic is distributed and where it is being sent.

Data plane – The underlying system helps in controlling traffic destination.

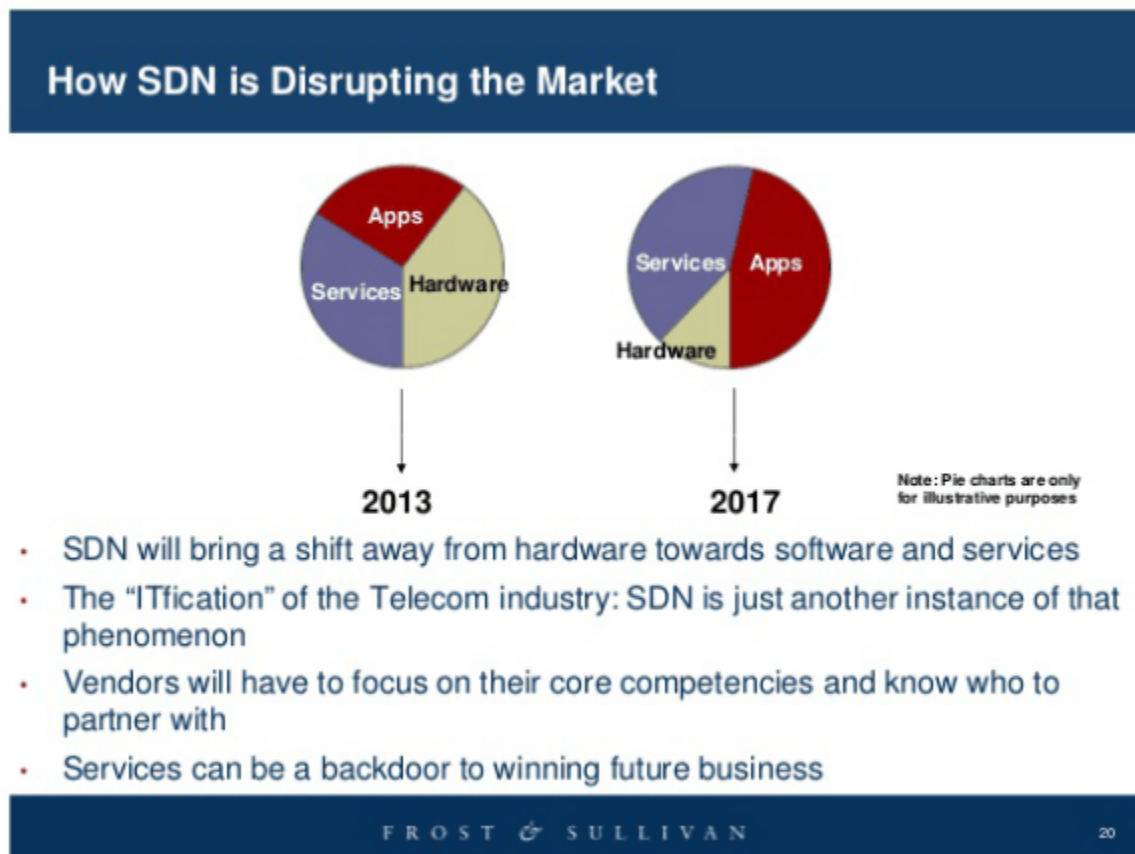
SDN application environment

Internal SDN Applications

Applications that are hosting the rest of the OpenDaylight controller software and are deployed internally, run inside the container. These applications must be written in the native language which is Java for ODL. Internal SDN applications must also adhere to the execution and design constraints of the controller. It must also execute in the same Java Machine as the controller which means that these types of the application must run locally with the controller. It can also access the MD-SAL applications and Java APIs of the controller running inside the controller's OSGi container.

External SDN Applications

Applications that are hosting the rest of the Open Daylight controller software, and are deployed externally, run outside the container. Any language can be used for writing External SDN applications that are scripting languages such as Bash. These applications can be run remotely which means on a different host than the controller. These applications will also use the application providing Restful access to their services and REST API provided by the controller.



Top Application and Service that can benefit from SDN are:

Security services

The modern virtualization ecosystem supports specific virtual service that is running within the network layer. It means an incorporating function like NFV into SDN platforms. This type of network security creates a genuinely proactive environment that is capable of risk reduction and responds to the incidents very quickly. Whenever a violation occurs, every second is quite critical to stop the attack. It is also essential to identify the attack and also to ensure that other network components are safe from the attack. As the organization in the modern era becomes even more digitized, and as the network layer becomes even more critical, we will see even more attacks and more advanced sophisticated advanced persistent threats. You will be able to create a more proactive environment that is capable of responding to the changes if you integrate potent services into the SDN layer.

Network Monitoring and Intelligence

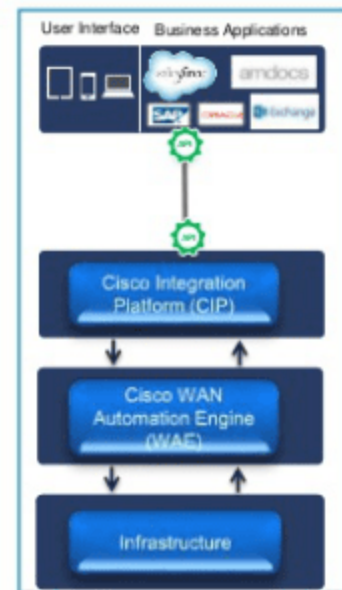
Modern SDN technologies help in abstracting one of the most critical layers within the data centre that is the network. Network architectures are very much complicated and have to handle a lot more data than ever before. This means it's critical to know what is following through your environment. Do you have remission issues on a port? What will happen if you are running heterogeneous network architecture? Or, are you passing a lot of traffic and are heavily virtualized through the network architecture? All of these challenges or issues are diminished if you have a solid network monitoring and intelligence layer. However, you also gain benefit and true insight if you integrate these technologies into your SDN architecture. Even optimization, alerting, hypervisor integration, port configurations, and traffic flow can be incorporated into network monitoring and intelligence technologies. Also, these types of agile systems will also help you to monitor network traffic between your cloud ecosystem and your data centre.

Bandwidth Management

With the help of SDN applications, operators can use bandwidth management to ensure the end users to receive online video watching and optimal browsing experiences. This SDN application can also monitor the bandwidth requirements then provision user flows to match the latency and bandwidth requirements of the layer 7 application. This type of application-aware approach to bandwidth management will also ensure a better user experience with zero buffering through better video playback. At this stage in the game, there is little doubt that SDN is becoming a reality in operator networks.

Bandwidth on Demand & Calendaring Application - SDN Application

- Web based Self Service Portal that enables Bandwidth on Demand provisioning with Calendaring support
- Benefits
 - Service Provider
 - Improved operational efficiency, faster time to market
 - New revenue opportunity
 - Cisco
 - Demonstrate Leadership in SDN applications
 - Evangelize and Accelerate SDN API adoption



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However, it is the SDN applications that will really bring powerful improvements to the operator's business and networks, beyond the immediate impact of simpler management of the network. And so the network infrastructure providers need to start mapping out this future to calculate all the potential that can be provided by SDN.

By acting and thinking ahead on SDN applications now, network infrastructure operators and providers will be able to rapidly evolve to provide flexible, customized networks that can entirely enhance their own bottom lines and can also enhance the end user experience.

Content Availability

There will be content servers used for media delivery or caching, on a service-provider edge network. These are installed by the content delivery network or operator service providers. Content that is to be served to the users is distributed and preoccupied across multiple content servers and also across various geographies in some cases.

SDN apps will be able to provision flows in the network based on the availability and types of the content which is built to handle content availability. SDN applications can also check the availability of the content in the content servers before routing requests to servers. A content-routing application will provide intelligence on its availability along with enabling discovery of content in the content servers.

This intelligence can be further used to route requests to the correct servers wherever the content is residing. Therefore, SDN application will direct requests from those websites which are non-cache-able and that generate active content to a server that provides active content rather than a caching server which significantly reduces network discontinuation.

Regulation and Compliance-Bound Applications

Major cloud vendors are now providing the capability to work and store with compliance and regulation-bound workloads. Now organizations have the option to extend architectures which have initially been very limited because of regulation into the cloud and distributed environments. How can you segment the traffic? How can you ensure that regulation and compliance workloads are persistently monitored and secured? Here SDN can be a great help for you.

Network points, network traffic travelling between switches, and even hypervisors can be controlled in SDN architecture. You should also remember that this layer abstracts virtual hardware and functions controls. This powerful layer can then span various virtualization points, locations, and even cloud locations.

High –Performance Applications

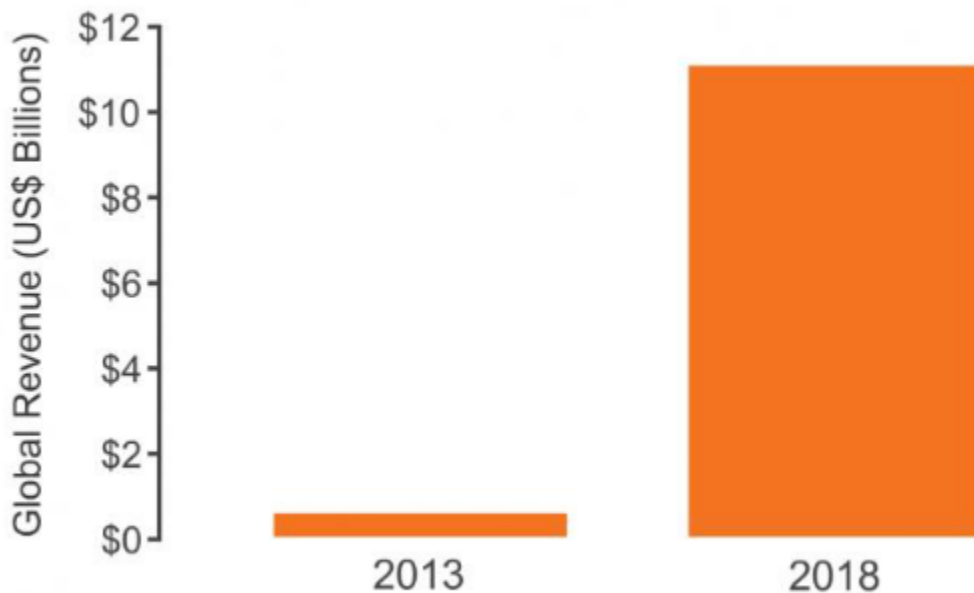
We are all seeing a rise in new types of application technologies. The delivery of rich apps like graphics design software, engineering, CAD, and GIS is allowed by virtualization. Traditionally, these workloads are required bare-metal architectures with their own connections. However, with the help of virtualization, VDI can help in creating powerful desktop experiences and applications are streamed. We can also see the integration of SDN into application control at the network layer. All of these functions like segmenting

heavy traffic, securing confidential data, creating powerful QoS policies, and even creating threshold alerts around bottlenecks within SDN will help to support rich and high-performance applications which are being delivered through virtualization.

Distributed Application Control and Cloud Integration

The capability to extend across the entire data centre is one of the most significant benefits of SDN. This type of agility integrates distributed cloud, locations and the organization as a whole. SDN also allows for critical network traffic to pass between various locations irrespective of the type of underlying network architecture. You also permit easier movement of data between cloud locations and data centre by abstracting critical network controls. You can utilise powerful APIs to not only integrate with a cloud provider, but you can also control specific network services as well because SDN is a form of network virtualization. While keeping your business agile, this allows you to manage your workloads granularly.

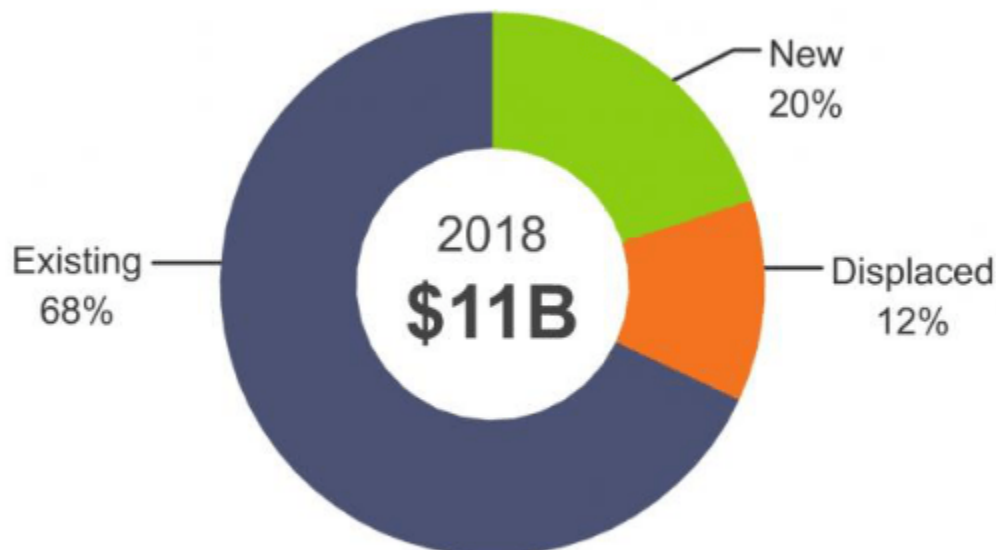
Infonetics expects carrier SDN and NFV hardware and software to grow from \$500M in 2013 to \$11B in 2018



© Infonetics Research, *Carrier SDN and NFV Hardware and Software: Annual Market Size and Forecasts*: November 2014

“For three years, the telecom industry has been abuzz over SDN and NFV, with anticipation and hard work developing the vision, goals, architectures, use cases, proof-of-concept projects, field trials, and even some commercial deployments. We’ve been gathering data in this early market for nearly two years and are projecting the global service provider SDN and NFV market to reach \$11 billion in 2018,” says [Michael Howard](#), Infonetics Research’s co-founder and principal analyst for carrier networks.

Global carrier SDN/NFV revenue will come from products that are new, displaced, and from existing market segments



© Infonetics Research, *Carrier SDN and NFV Hardware and Software: Annual Market Size and Forecasts*: November 2014

SDN applications are being used by organizations for a lot of the functions; however, these were few of the main features to consider. You should understand how **SDN applications** can positively impact your business and your data centre. SDN fundamentally simplifies the entire networking layer and provides you granular control over your distributed data centre ecosystem, services, and applications.

Also, SDN helps you to design a business capable of adjusting to changes in the industry and market shifts. This also allows your organization to be truly productive and agile.